

Welcome to QuACK!

Week 8
10/29/2026
Nina & Alyson

Today's Agenda

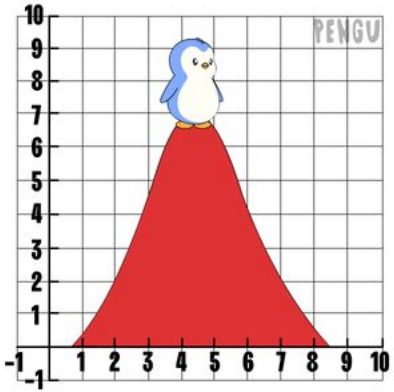
- Warm-up activity
- Analyzing your data
  - Goals
  - Methods
  - Steps
- Group Activity
- Demo: Data Analysis



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Week 8 Warm-up - 10 Minutes

- Download this week's materials from the website. (GradLife_SemWide8)
- Find the number of respondents in the dataset and save it to a variable "n"
- Find the mean, median, and range of self-care at the final time (S3)
- Make a boxplot comparing caffeine usage (S1) for respondents in Cognition vs. Social.



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What is the point of analyzing your data?

→ to know to what degree you should believe that any patterns you found in your data are representative of the underlying population.



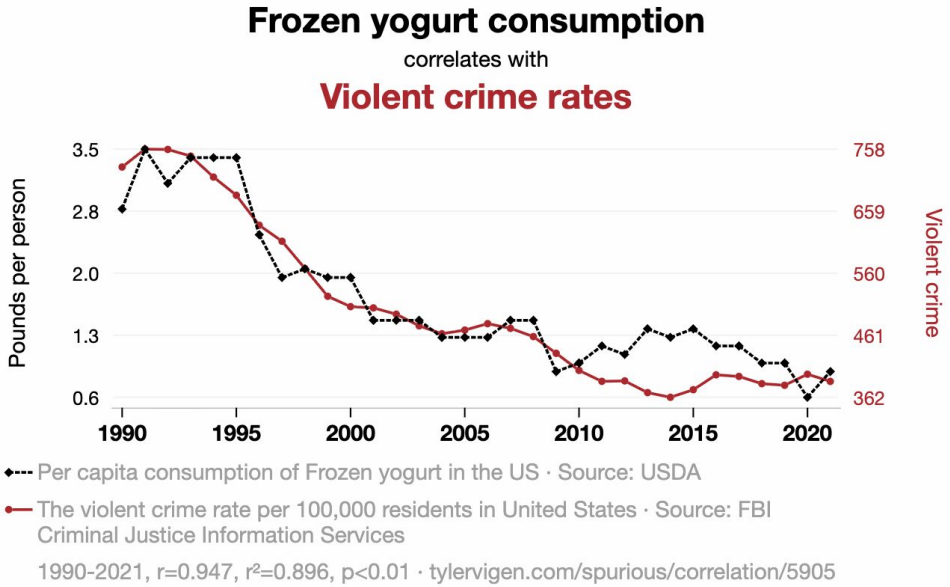
How do you decide how to analyze your data?

- Tricky because there is no one-size-fits-all answer
- For common types of datasets (e.g., comparing means of numeric variables across two groups of equal size), there are well-established tests
- For others (e.g., time-series data or analyses on especially small datasets), you may need some more niche methods
- We’re going to focus on building up intuitions and R skills for basic statistical tests

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Correlation vs. Causation

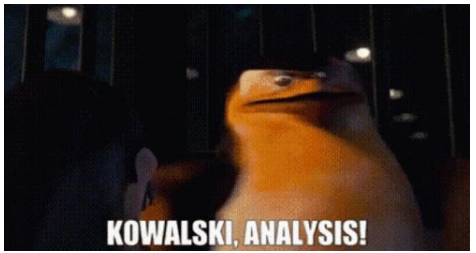
- **Correlation**: a directional relationship between two variables
- **Causation**: a change in one variable can be directly linked to a change in another variable



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Basic steps to analyzing your data
(some of these steps happen long before you even look at your data!)

- Identify your independent and dependent variables (and groups/levels)
- Decide what your null and alternative hypotheses are (H_0 and H_a)
- Pick your test, α, and do a power analysis
- Calculate your test statistics (e.g., t for t-tests or F for ANOVAs)
- Decide whether to retain or reject H_0
- Make sure you can clearly explain your conclusion in words!



Activity: What kind of analysis can we use?

In our dataset (GradLife), what statistical test would you use to:

1. Determine whether self-care at the end of the study (S3) is better in Developmental or Clinical?
2. Determine whether caffeine intake at the beginning (S1) is related to TikTok usage at the beginning (S1)?
3. Compare self-care scores (S3) across multiple areas?
4. Predict hours contemplating other paths (S3) from self-care (S3) and submission of grant status?
5. Predict grant submission from TikTok usage (S2) and caffeine consumption (S2) ?

Example Scenario

You are conducting a series of studies on children's attention based on their screen time. You will manipulate the type of content that children are consuming and assess their attention at multiple timepoints.

Dependent Variables:

- Attention (lookaway time during picture book reading) at pre-intervention, midway, and post-intervention timepoints

Independent Variables:

- Location (rural, urban)
- Age (continuous)
- Screen Time (continuous)
- Content Type (education only, entertainment only, combination)

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Overview of basic statistical tests

Are you interested in **relationships between variables** or **comparing means**?

Relationships between Variables
Do you want to just see if there is a relation or predict one variable based on another?

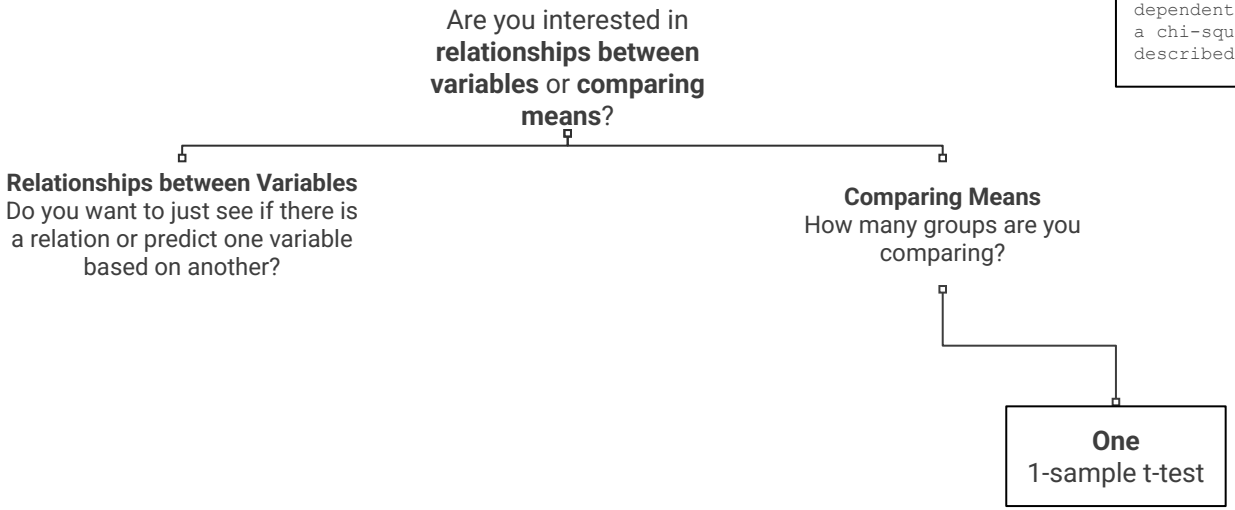
Comparing Means
How many groups are you comparing?

Note: the specific test you choose will also depend on the type of variables you have (continuous, categorical, count, etc.). For example, if you have categorical independent and dependent variables, you may need a chi-square test which is not described here.

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Overview of basic statistical tests

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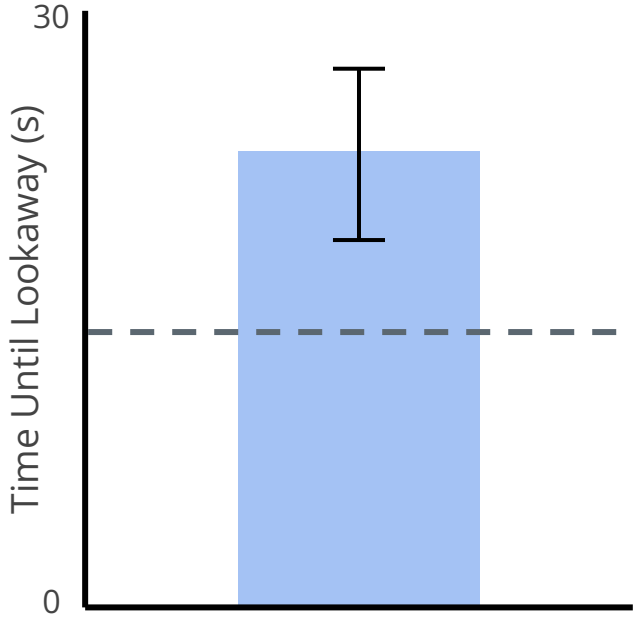
1-sample t-test example

A previously published study exploring children's lookaway time during a picture book reading task found that children on average looked away from a page after 15 seconds.

You want to see how your data compare to this previous study.

How many groups? **One**

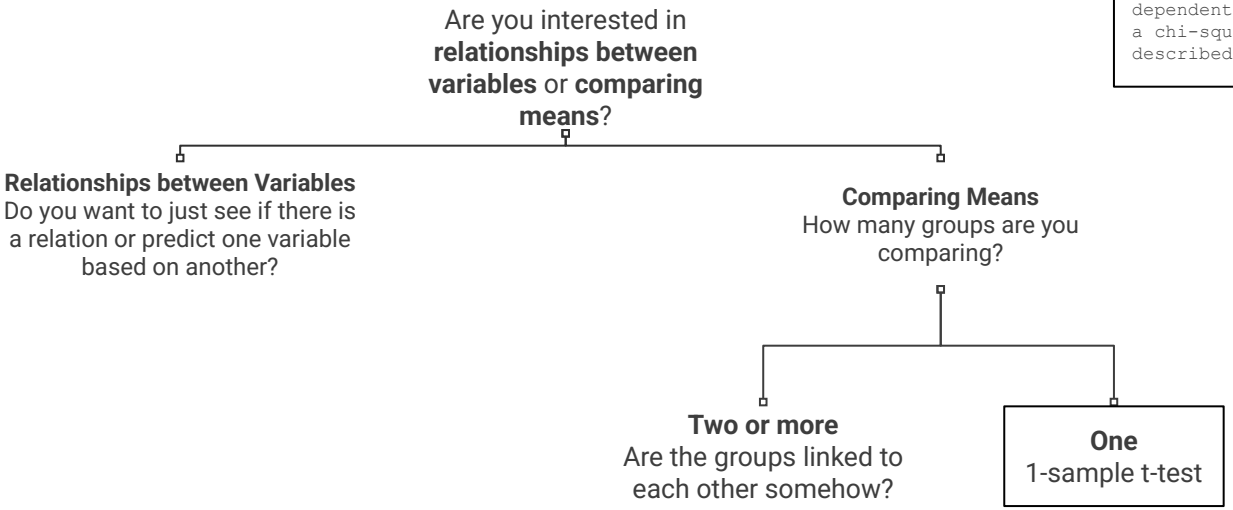
- Mean attention averaged across all participants in your data compared to mean from previous study (15s)



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Overview of basic statistical tests

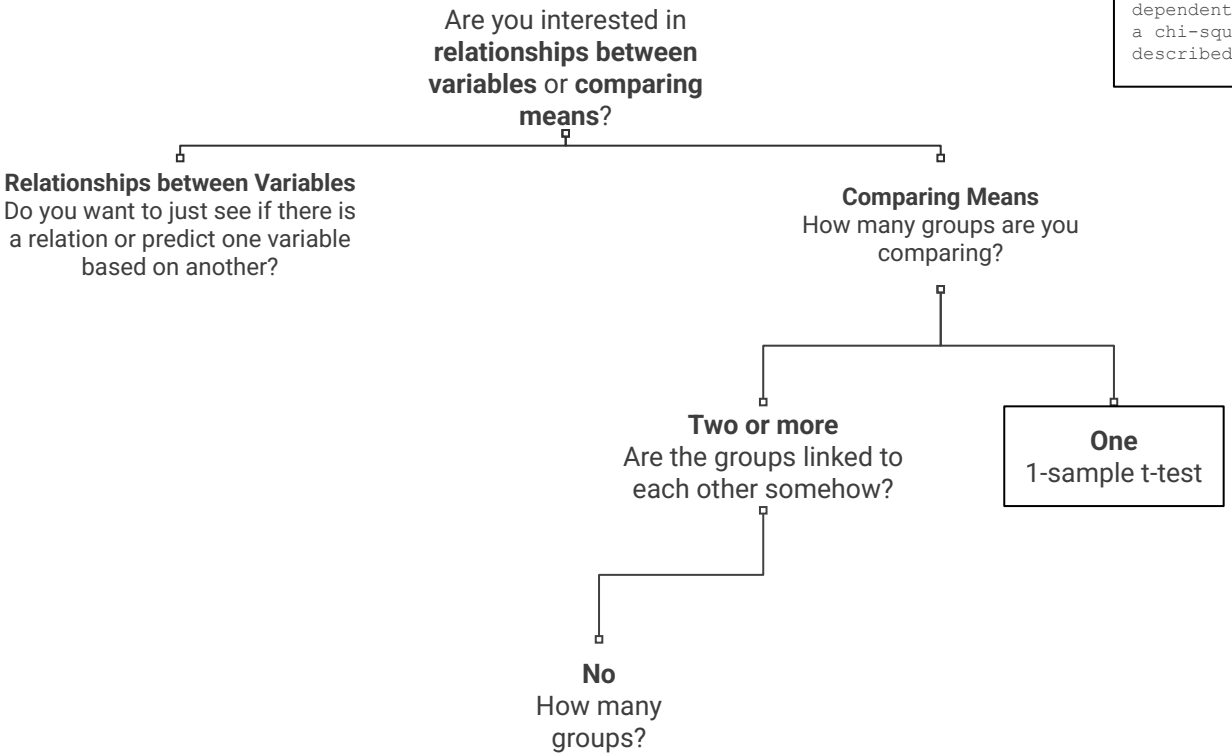
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Overview of basic statistical tests

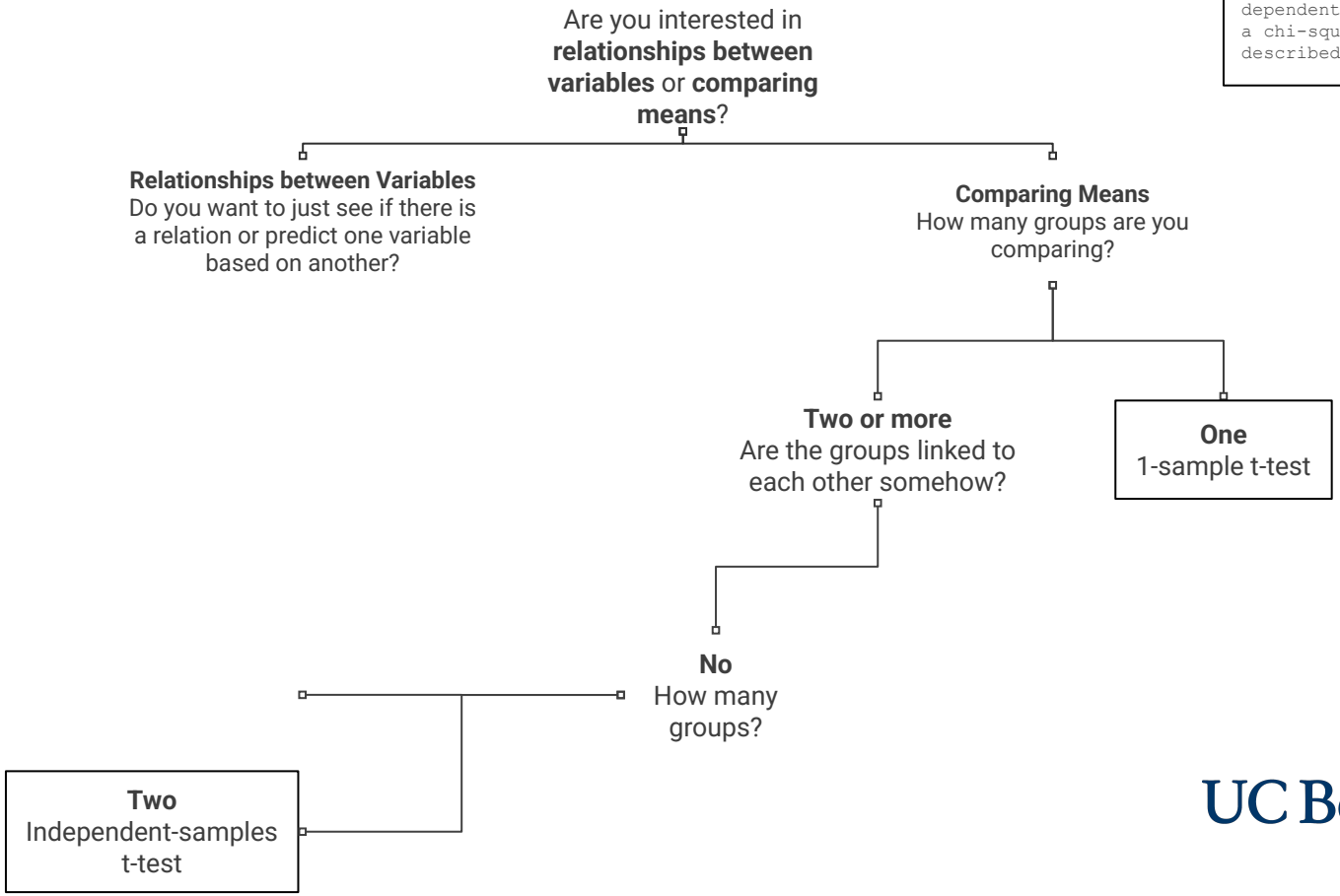
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Independent-samples t-test example

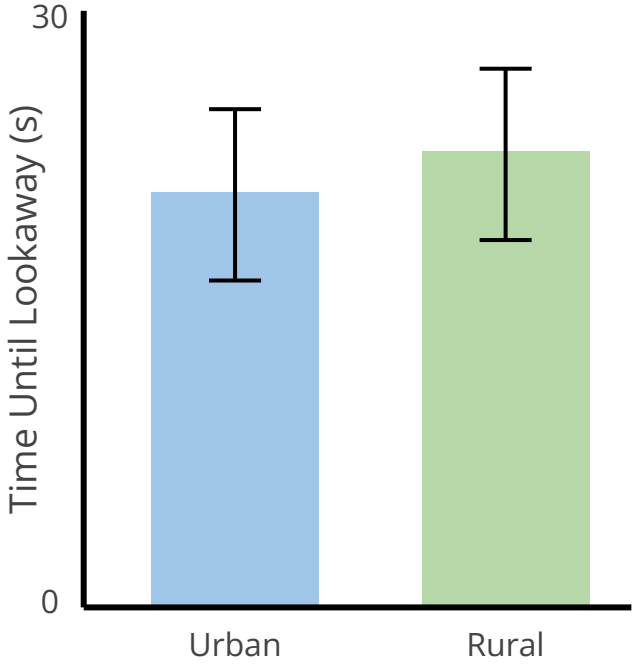
Now, you want to explore whether there was a difference in the mean attention between children who lived in rural communities vs. urban communities.

How many groups? **Two**

- Rural
- Urban

Are the groups linked? **No**

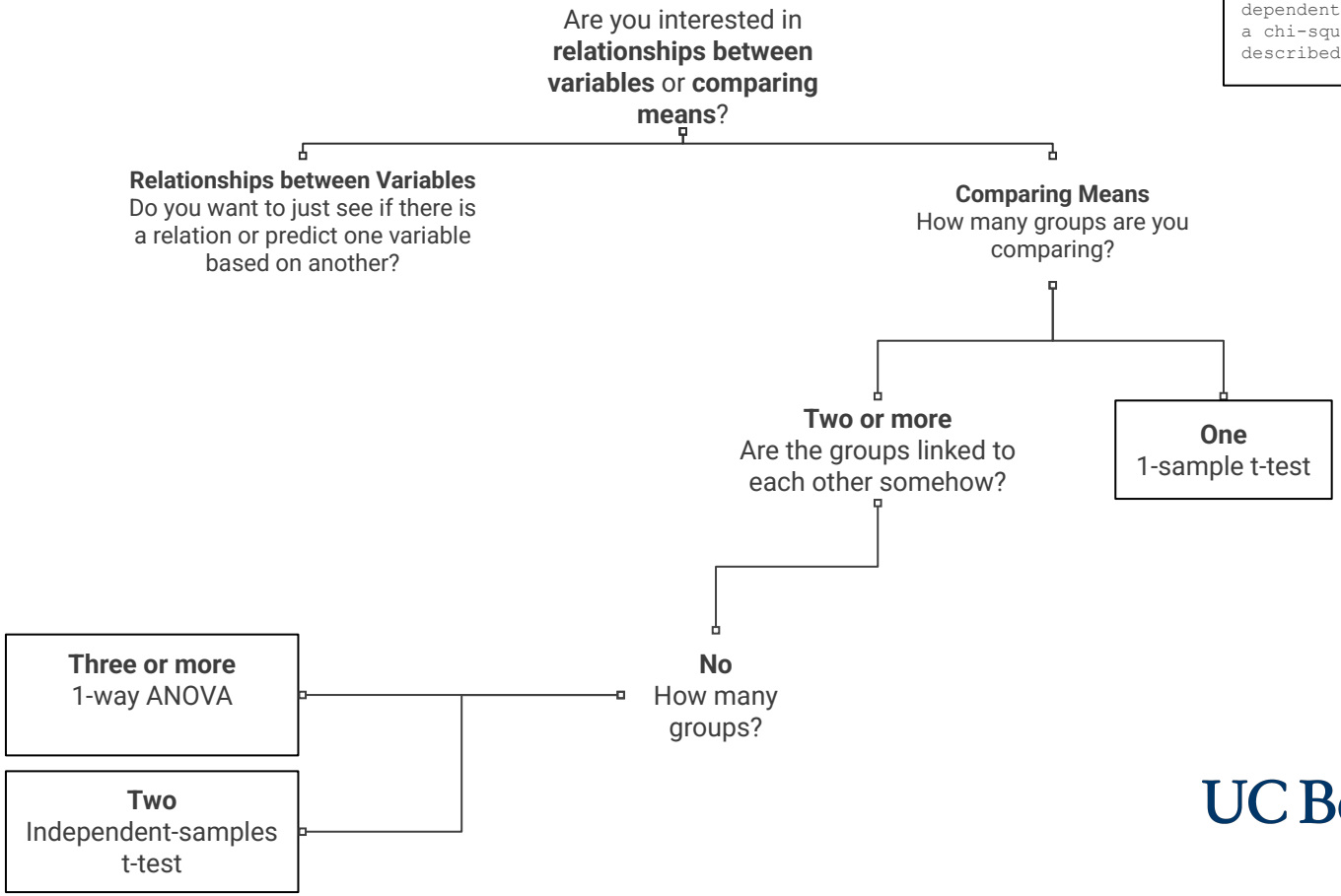
- Different children in each group



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Overview of basic statistical tests

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1-way ANOVA example

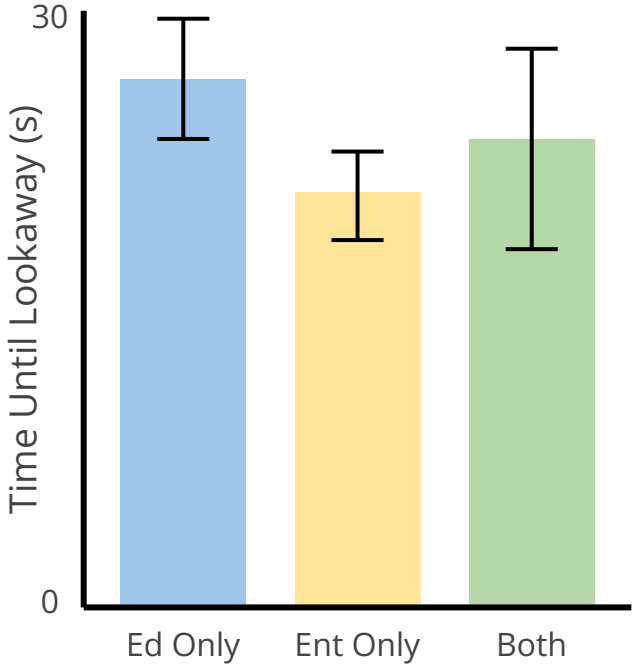
Next, you want to explore whether there was a difference in the mean attention based on the type of content children were exposed to (educational only, entertainment only, or both).

How many groups? **Three**

- Educational Only
- Entertainment Only
- Both

Are the groups linked? **No**

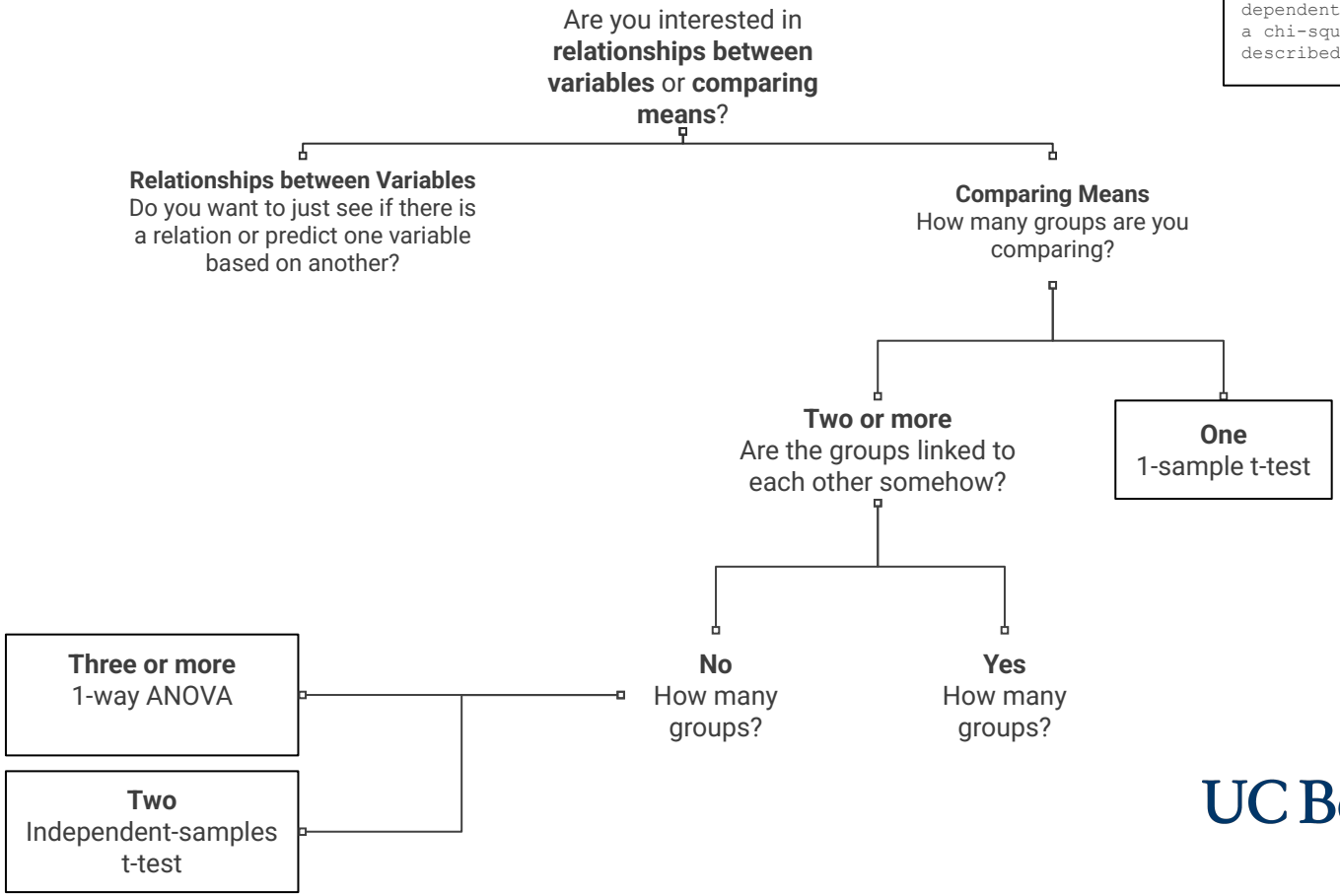
- Different children in each group



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Overview of basic statistical tests

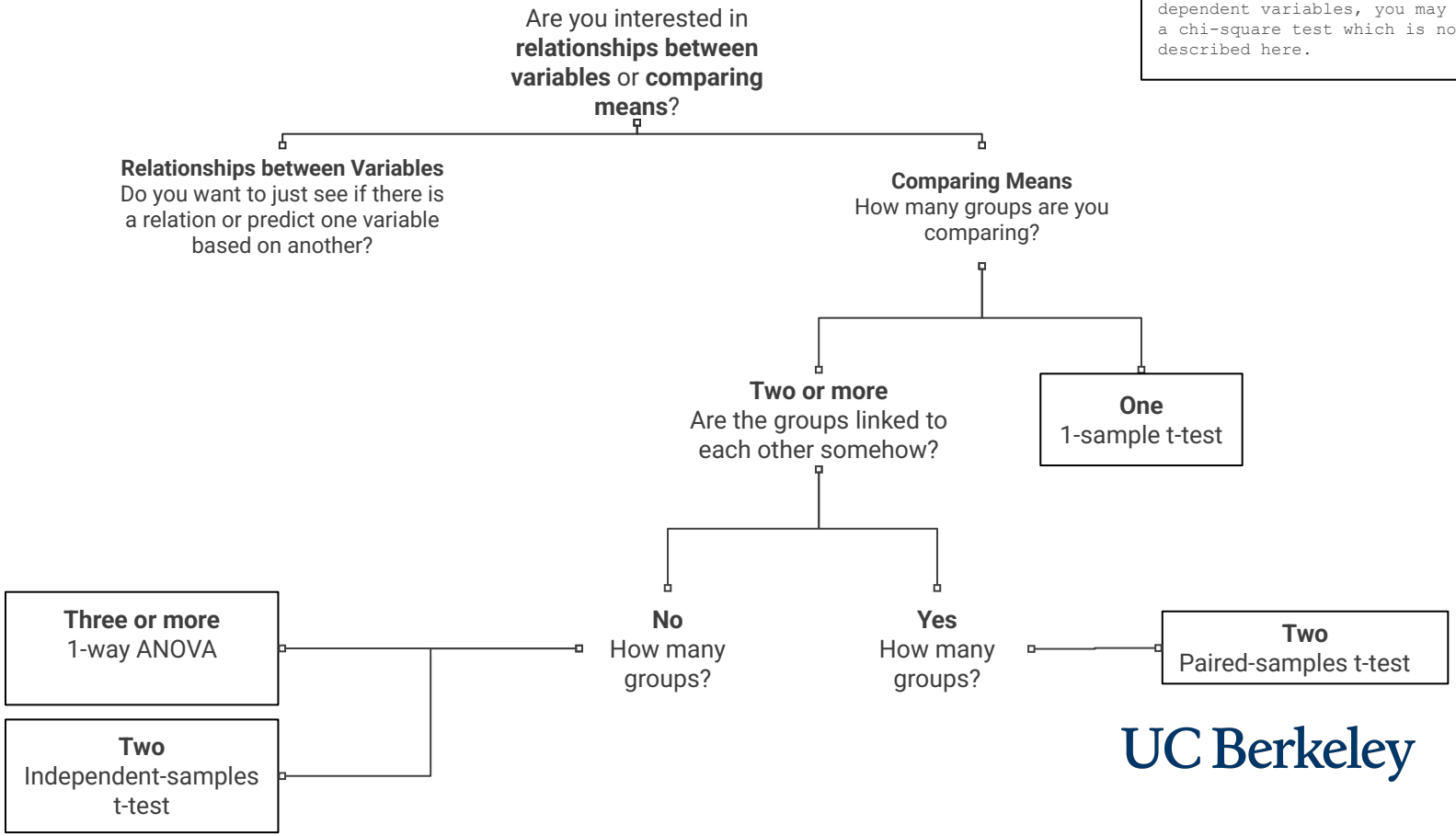
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Paired-samples t-test

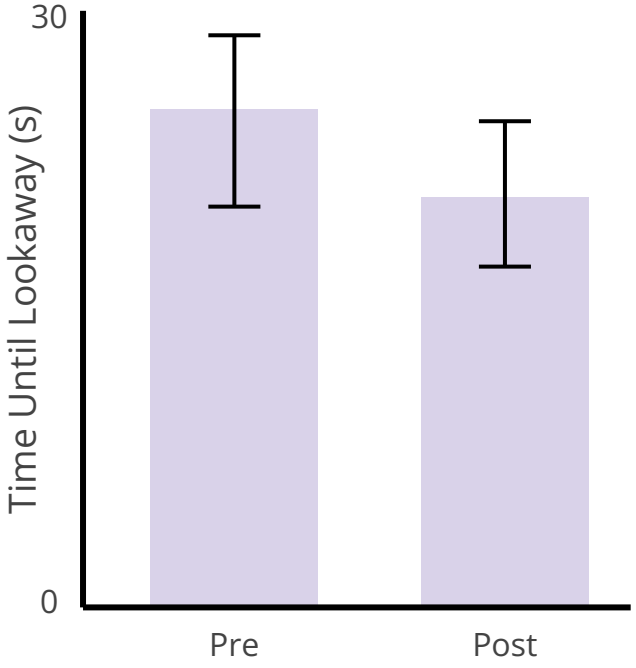
You now want to compare Pre-Intervention mean attention vs. Post-Intervention mean attention.

How many groups? **Two**

- Pre-Intervention
- Post-Intervention

Are the groups linked? **Yes**

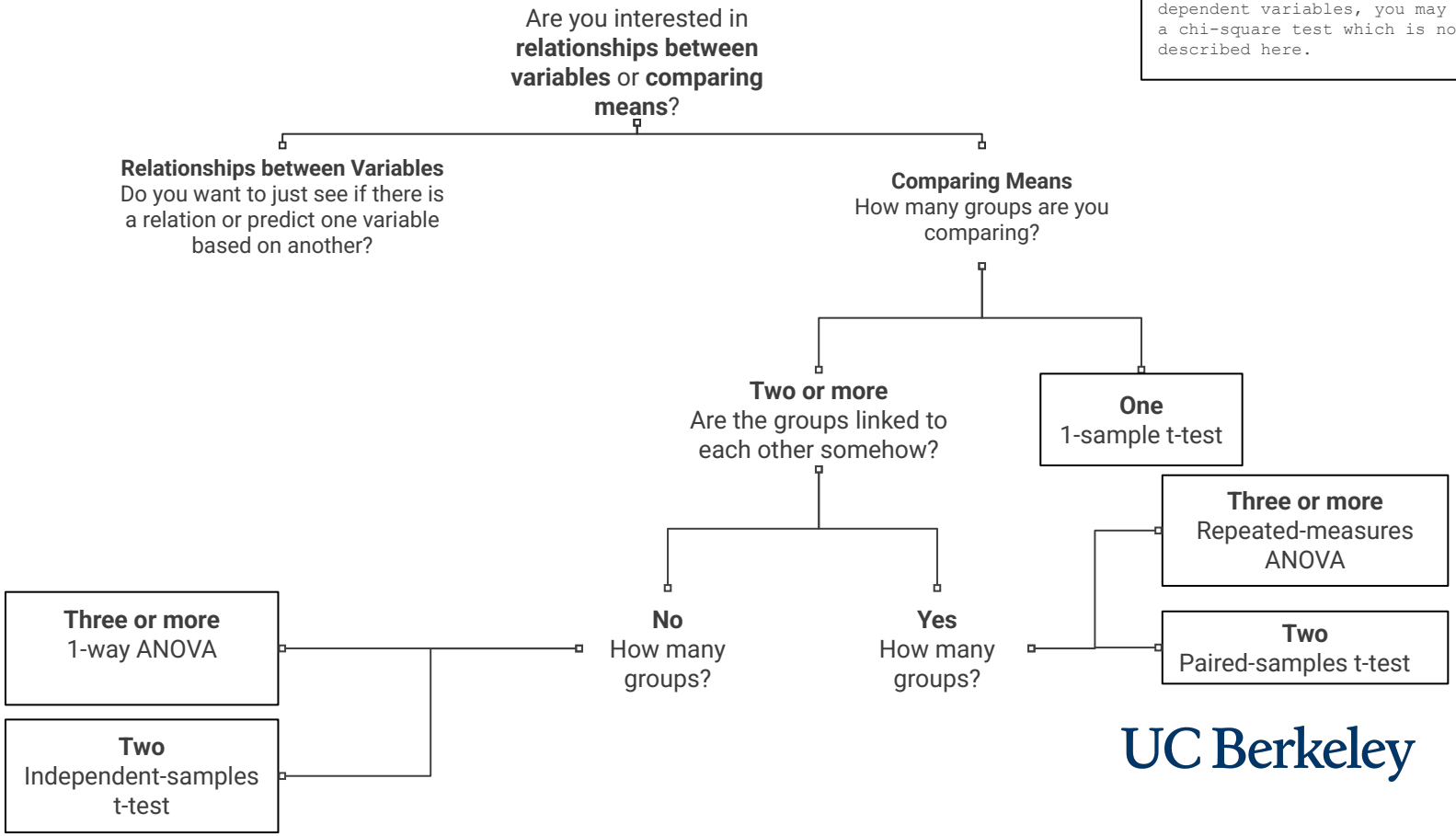
- The same children are in the pre-intervention and post-intervention groups (same children being measured at different time points)



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Overview of basic statistical tests

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Repeated measures ANOVA

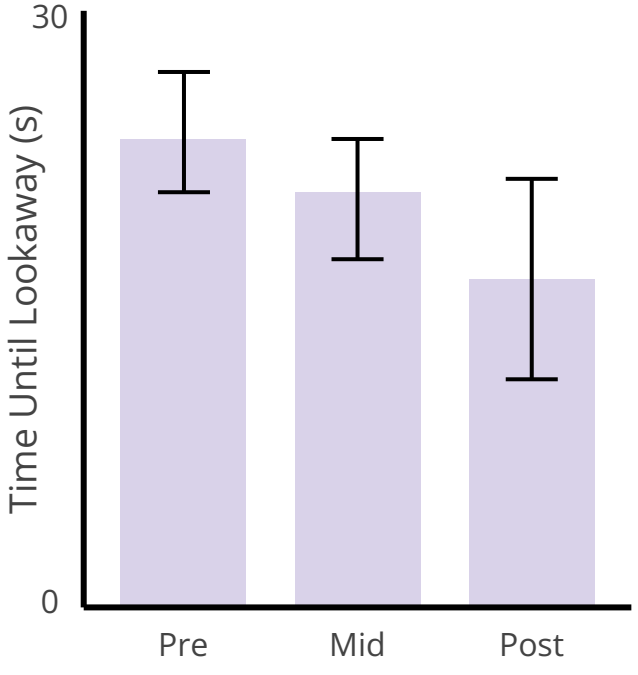
Comparing mean attention at all time points (pre-intervention, mid-point, and post-intervention).

How many groups? **Three**

- Pre-intervention
- Mid-point
- Post-intervention

Are the groups linked? **Yes**

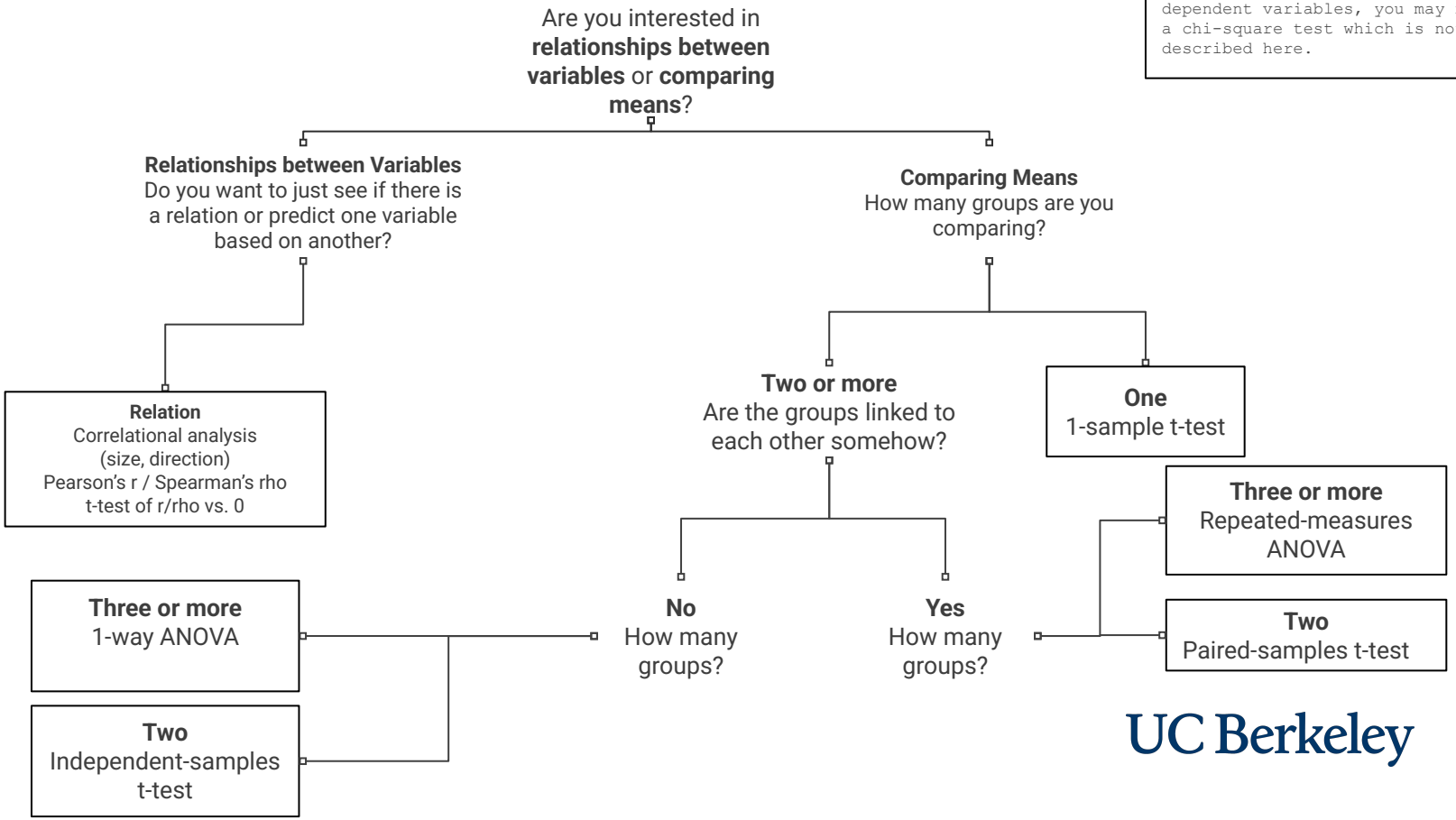
- Same children measured at different time points



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Overview of basic statistical tests

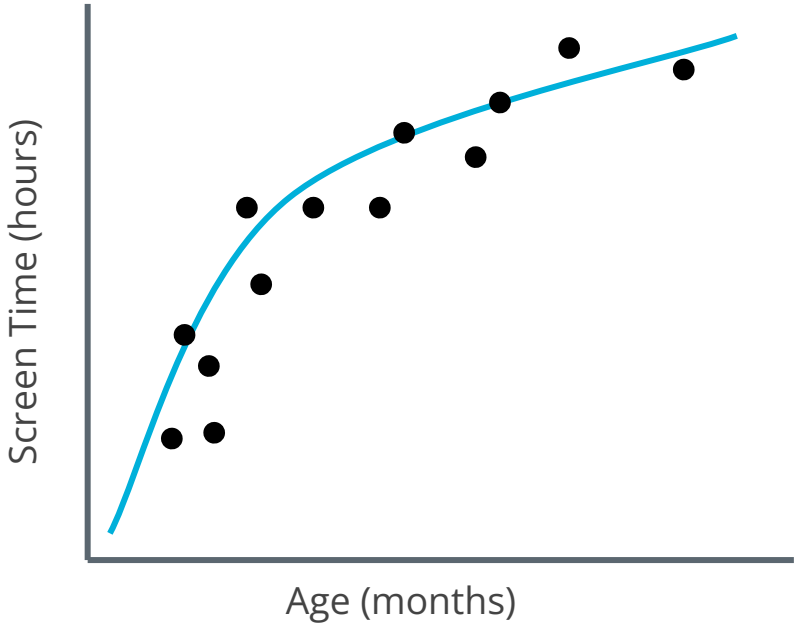
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Correlation Analysis

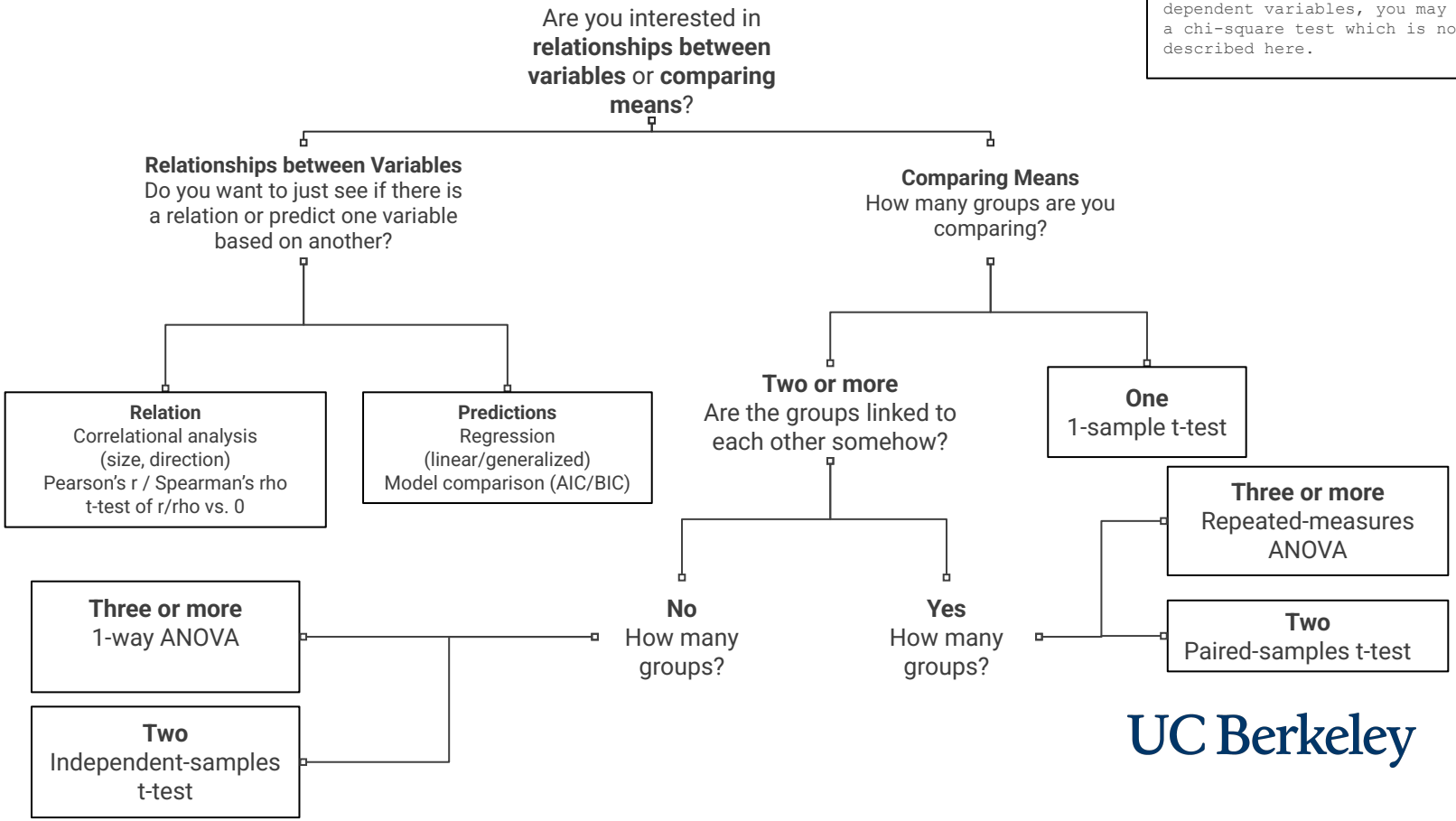
Age in months and screen time in hours



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Overview of basic statistical tests

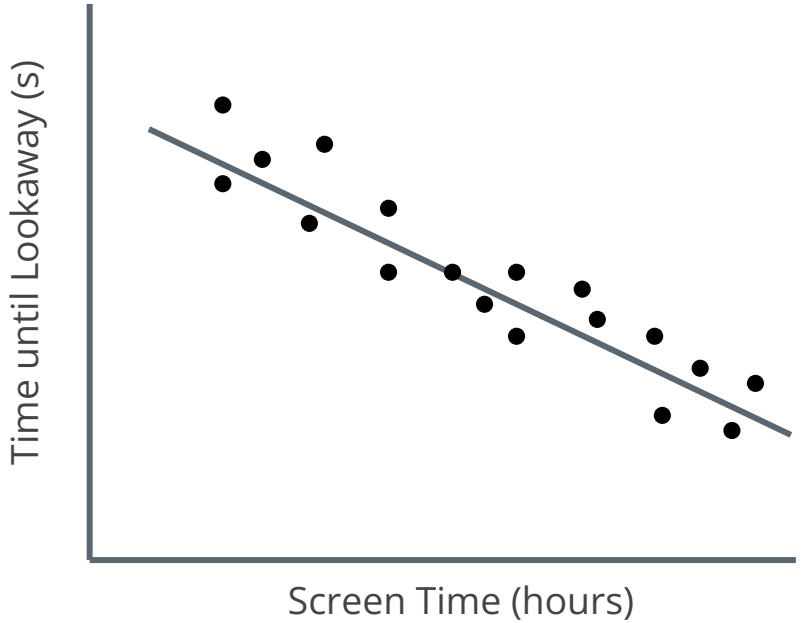
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Regression / Model Comparison

Mean attention predicted by screen time



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And as usual, resources you can look back on even when QuACK is over 🥲

https://www.statsflowchart.co.uk/

quiz that will tell you which statistical test to use

https://stats.oarc.ucla.edu/other/mult-pkg/whatstat/

table to help you figure out which statistical test to use + how to do in SAS, Stata, SPSS, R

https://statsandr.com/blog/what-statistical-test-should-i-do/

decision tree with explanations of different statistical tests

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Exit Ticket

Link: https://tinyurl.com/quack-fall2026-exit-survey



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